

LABORATORY OBSERVATION

Course : Full Stack Web Development

Course Code: 23CM4121

Experiment No: 5

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1. TITLE

Dynamic Webpage Using DOM in JavaScript – Café Menu

2. AIM

To understand the Document Object Model (DOM) and demonstrate how JavaScript dynamically changes the content, style, and attributes of HTML elements in a café menu webpage.

3. OBJECTIVE

- To understand the structure and purpose of the Document Object Model (DOM).
 - To access HTML elements using `document.getElementById()`.
 - To dynamically change HTML content using `innerHTML`.
 - To dynamically change the style of HTML elements using JavaScript.
 - To change HTML attributes using `setAttribute()`.
 - To change the webpage theme using `classList.toggle()`.
 - To create an interactive and dynamic webpage.
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4. THEORY

The Document Object Model (DOM) is a programming interface that represents an HTML document as a tree of objects or nodes. JavaScript uses the DOM to access, modify, add, or remove HTML elements and their properties.

When a webpage is loaded, the browser creates a DOM tree from the HTML code. JavaScript can select an element using methods such as `document.getElementById()` and then change its content, style, or attributes.

In this application, the DOM is used to create an interactive café menu. Buttons are used to change the café title, special item, description, coffee icon style, and page theme.

The main DOM methods and properties used are:

- `document.getElementById()` – selects an HTML element using its id.
- `innerHTML` – changes the visible text content of an element.
- `element.style` – changes the inline CSS style of an element.
- `setAttribute()` – changes or adds an HTML attribute.

- `classList.toggle()` – adds or removes a CSS class.
- `onclick` – executes a JavaScript function when a button is clicked.

Basic flow: User Clicks Button → JavaScript Function Executes → DOM Element Is Selected → Content / Style / Attribute Changes → Updated Webpage Is Displayed

5. ALGORITHM / PROCEDURE

1. Create the HTML structure for the café menu webpage.
 2. Add CSS styles for the body, menu card, coffee icon, buttons, and dark theme.
 3. Create HTML elements with unique id values such as title, special, description, coffee, and status.
 4. Create buttons and connect each button to a JavaScript function using `onclick`.
 5. Use `document.getElementById()` to select the required HTML element.
 6. Use `innerText` to change the title, special item, or description.
 7. Use style properties to change colors, font size, font style, background, border, and transform.
 8. Use `setAttribute()` to change the title attribute of the coffee icon.
 9. Use `classList.toggle()` to switch between the normal and dark themes.
 10. Update the status message after each operation.
 11. Run the HTML file in a web browser and test all buttons.
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6. PROGRAM

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>DOM Cafe Menu</title>
  <style>
    * {
      box-sizing: border-box;
    }
    body {
      margin: 0;
      min-height: 100vh;
      font-family: Arial, sans-serif;
      background: #f6efe5;
      display: flex;
      justify-content: center;
      align-items: center;
      transition: 0.4s;
    }
    .menu-card {
      width: 450px;
      background: #fffaf3;
      padding: 30px;
      border-radius: 20px;
      text-align: center;
      box-shadow: 0 10px 30px rgba(70, 45, 25, 0.18);
      transition: 0.3s;
    }
    .menu-card:hover {
      transform: translateY(-5px);
    }
    .coffee-icon {
      width: 110px;
      height: 110px;
      margin: auto;
      border-radius: 50%;
```

```

        background: #d89b6c;
        border: 10px solid #754c32;
        position: relative;
        box-shadow: 0 5px 15px rgba(60, 40, 25, 0.2);
    }
    .coffee-icon::before {
        content: "";
        position: absolute;
        width: 45px;
        height: 45px;
        background: #5a3825;
        border-radius: 50%;
        left: 22px;
        top: 22px;
    }
    .coffee-icon::after {
        content: "";
        position: absolute;
        width: 25px;
        height: 25px;
        border: 5px solid #754c32;
        border-radius: 50%;
        right: -30px;
        top: 28px;
    }
    h1 {
        margin: 18px 0 5px;
        color: #4b3022;
    }
    #special {
        color: #a65f32;
        font-weight: bold;
    }
    #description {
        color: #75675d;
        line-height: 1.6;
    }
    .buttons {
        margin-top: 25px;
        display: flex;
        flex-wrap: wrap;
        justify-content: center;
        gap: 8px;
    }
    button {
        border: none;
        padding: 10px 14px;
        border-radius: 8px;
        background: #9b5b35;
        color: white;
        cursor: pointer;
        transition: 0.2s;
    }
    button:hover {
        background: #704025;
        transform: scale(1.04);
    }
    .status {
        margin-top: 20px;
        padding: 10px;
        border-radius: 8px;
        background: #f0e3d4;
        color: #665044;
        font-size: 14px;
    }
    .dark {
        background: #29231f;
    }
    .dark .menu-card {

```

```

        background: #3b322d;
    }
    .dark h1 {
        color: #fff7ee;
    }
    .dark #description {
        color: #d8cec4;
    }
    .dark .status {
        background: #50453e;
        color: #eee3d8;
    }
    .dark #special {
        color: #e5a475;
    }
</style>
</head>
<body>
    <div class="menu-card">
        <div class="coffee-icon" id="coffee" title="Coffee cup"></div>
        <h1 id="title">Cozy Corner Café</h1>
        <div id="special">Today's Special: Cappuccino</div>
        <p id="description">
            Enjoy freshly prepared coffee and delicious
            snacks in a warm and relaxing atmosphere.
        </p>
        <div class="buttons">
            <button onclick="changeTitle()">Change Title</button>
            <button onclick="changeSpecial()">Change Special</button>
            <button onclick="changeDescription()">Change Description</button>
            <button onclick="changeCoffee()">Change Coffee</button>
            <button onclick="changeTheme()">Change Theme</button>
        </div>
        <div class="status" id="status">
            DOM is ready. Try the buttons!
        </div>
    </div>

    <script>
        // Change HTML content
        function changeTitle() {
            const title = document.getElementById("title");
            title.innerText = "Welcome to Brew & Bean!";
            document.getElementById("status").innerText =
                "Cafe title changed using the DOM.";
        }

        // Change content and style
        function changeSpecial() {
            const special = document.getElementById("special");
            special.innerText =
                "Today's Special: Chocolate Mocha";
            special.style.color = "#7b3f20";
            special.style.fontSize = "18px";
            document.getElementById("status").innerText =
                "Special item and style changed dynamically.";
        }

        // Change content and style
        function changeDescription() {
            const description =
                document.getElementById("description");
            description.innerText =
                "Fresh coffee, tasty desserts and a perfect place to relax.";
            description.style.fontSize = "17px";
            description.style.fontStyle = "italic";
            document.getElementById("status").innerText =
                "Description and font style changed.";
        }
    </script>

```

```

// Change style and attribute
function changeCoffee() {
    const coffee = document.getElementById("coffee");
    coffee.style.background = "#d6a6e8";
    coffee.style.borderColor = "#814b9b";
    coffee.style.transform = "rotate(8deg) scale(1.08)";
    coffee.setAttribute("title", "Updated coffee cup");
    document.getElementById("status").innerText =
        "Coffee style and title attribute changed.";
}

// Change page theme
function changeTheme() {
    document.body.classList.toggle("dark");
    document.getElementById("status").innerText =
        "Page theme changed using the DOM.";
}
}
</script>
</body>
</html>

```

7. CODE EXPLANATION

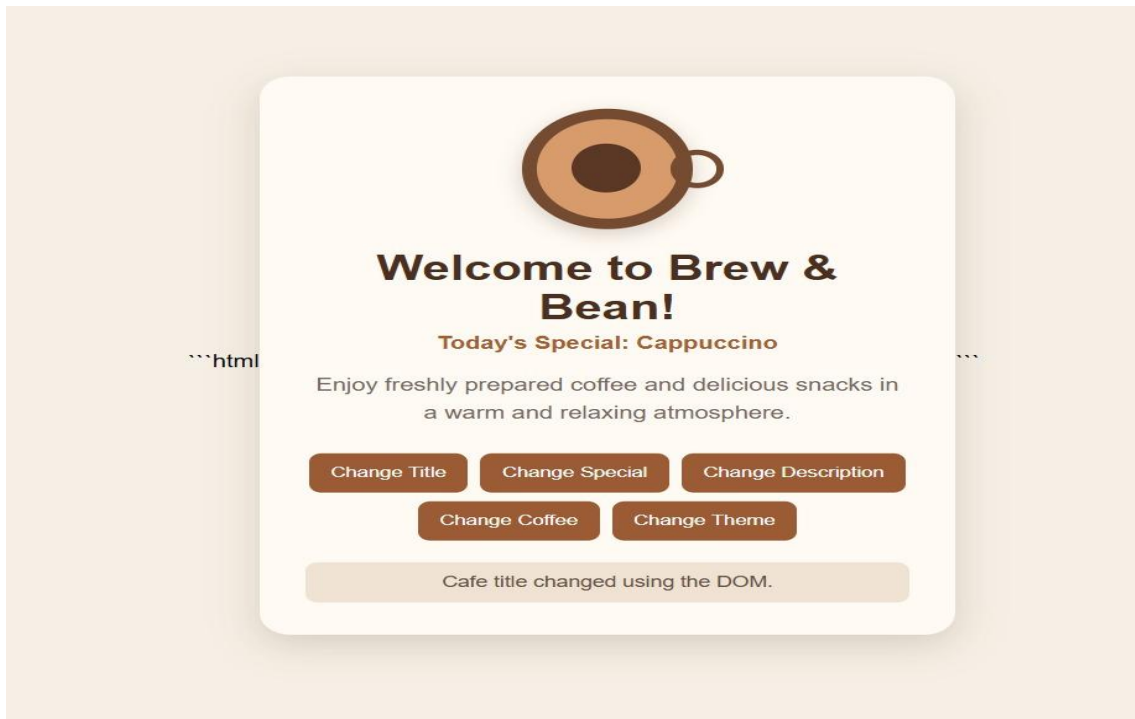
Code / Function	Explanation
document.getElementById()	Selects an HTML element using its id.
innerText	Changes the visible text inside an HTML element.
element.style.color	Changes the text color dynamically.
element.style.fontSize	Changes the font size of an element.
element.style.fontStyle	Changes the font style, such as italic.
element.style.background	Changes the background color of the coffee icon.
element.style.transform	Rotates and scales the coffee icon.
setAttribute()	Changes the title attribute of the coffee icon.
classList.toggle()	Adds or removes the dark class to change the theme.
onclick	Calls a JavaScript function when a button is clicked.
console-free DOM update	The changes are directly reflected on the webpage without reloading it.

8. TEST CASES AND EXECUTION DATA

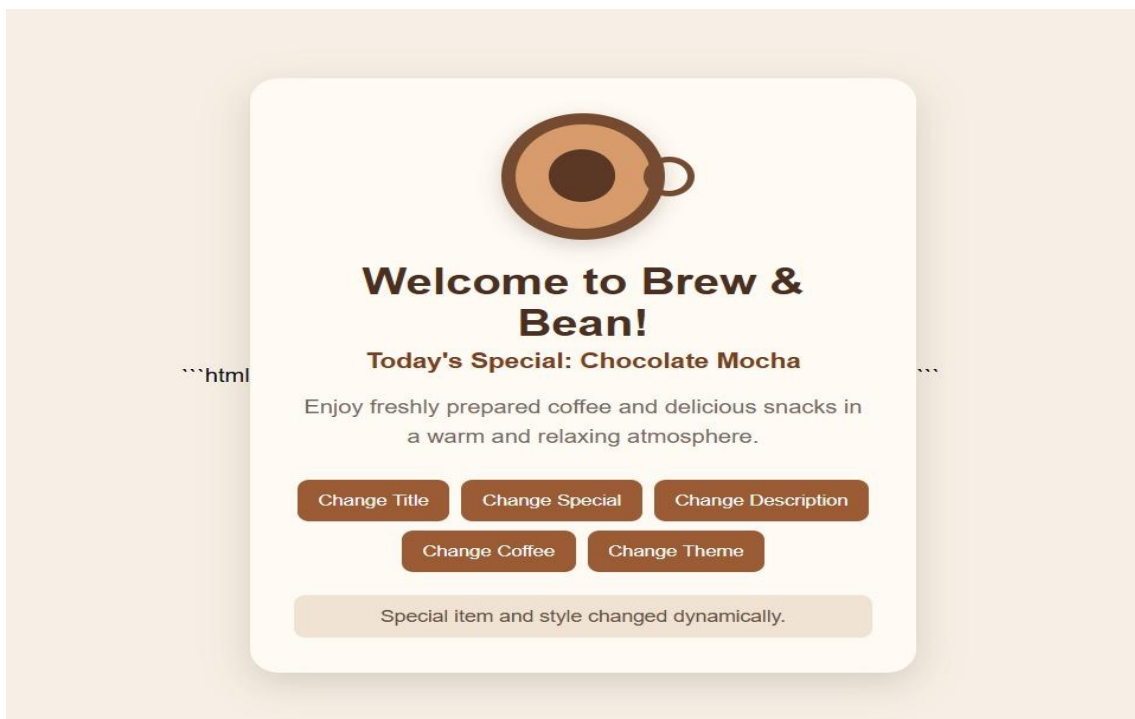
Test Case	Action	Expected Result	Status
TC-01	Click Change Title	Title changes to "Welcome to Brew & Bean!"	Pass
TC-02	Click Change Special	Special changes to Chocolate Mocha with new style	Pass
TC-03	Click Change Description	Description changes and becomes italic	Pass
TC-04	Click Change Coffee	Coffee icon color, border, rotation, scale, and title change	Pass
TC-05	Click Change Theme	Dark theme is applied or removed	Pass

9. OUTPUT

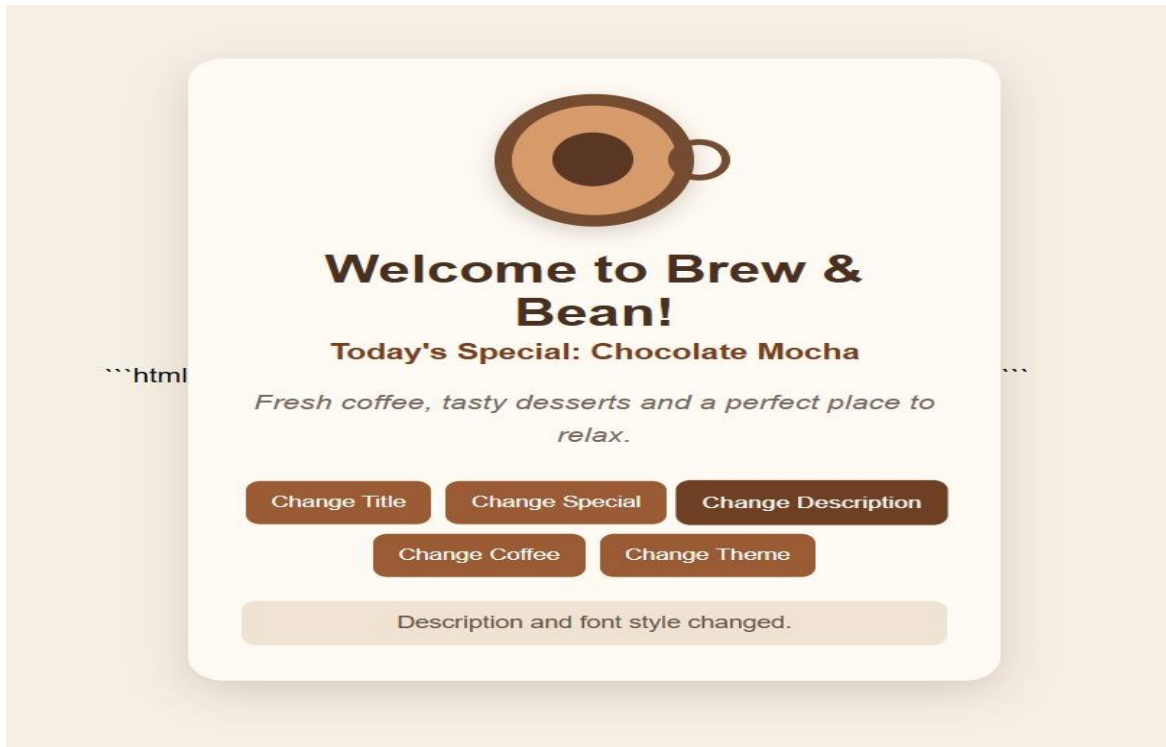
TC-01:



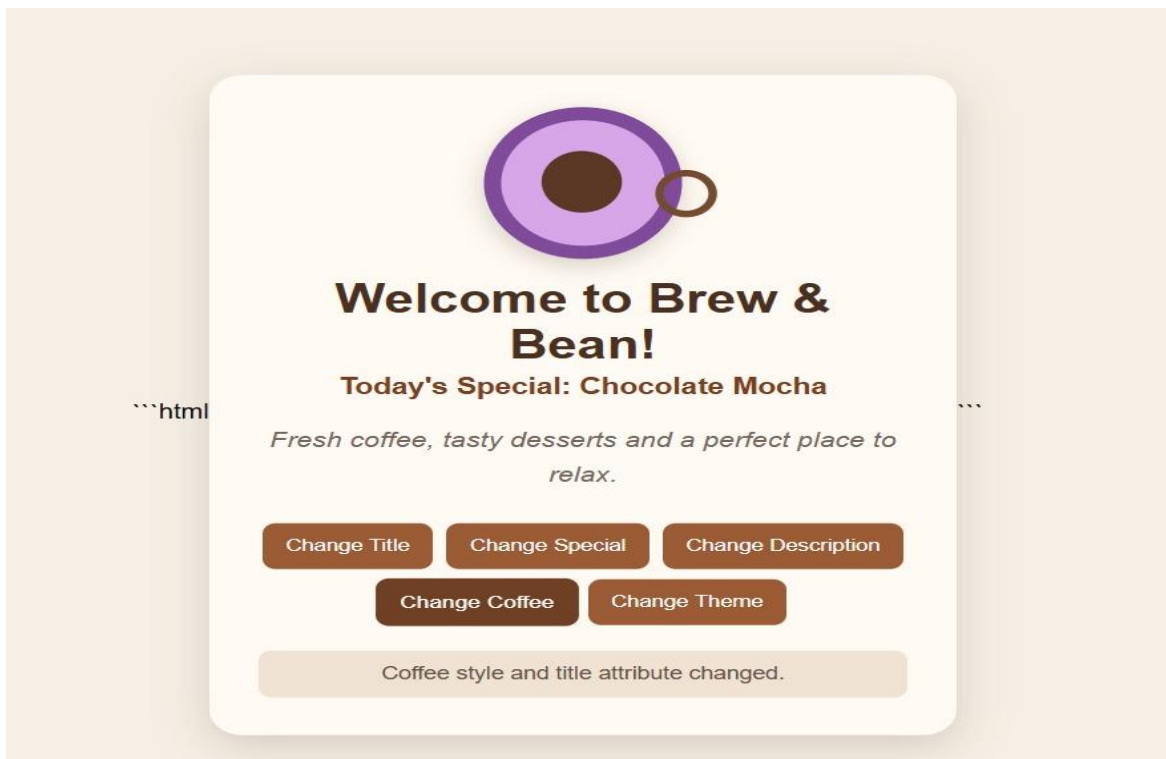
TC-02:



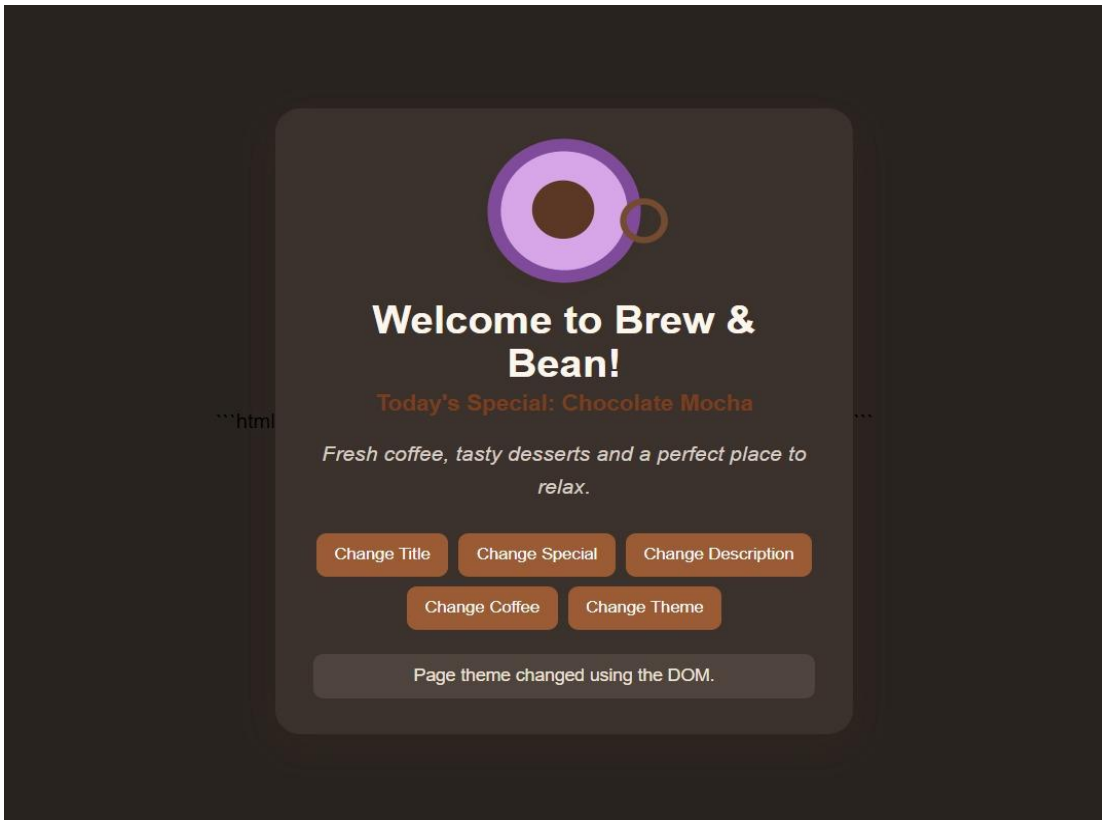
TC-03:



TC-04:



TC-05:



10. RESULT

Thus, a dynamic café menu webpage was successfully developed using HTML, CSS, and JavaScript DOM methods. The application changes HTML content, styles, attributes, and the webpage theme dynamically when the user clicks the buttons.